

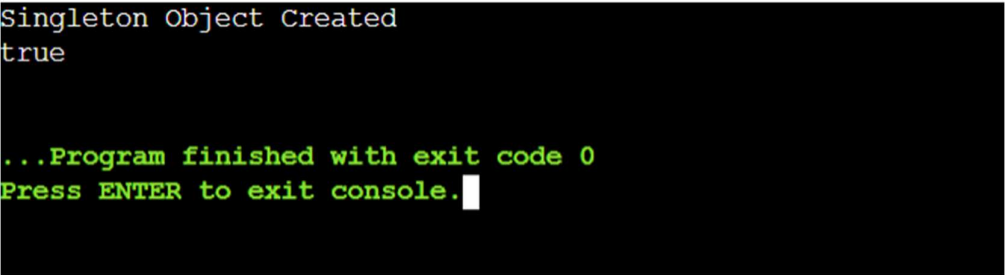
Week 1

Design Patterns and Principles

Exercise 1: Implementing the Singleton Pattern

//java code

```
class Singleton {  
    private static Singleton obj;  
  
    private Singleton() {  
        System.out.println("Singleton Object Created");  
    }  
  
    public static Singleton getInstance() {  
        if (obj == null) {  
            obj = new Singleton();  
        }  
        return obj;  
    }  
}  
  
public class Main {  
    public static void main(String[] args) {  
        Singleton s1 = Singleton.getInstance();  
        Singleton s2 = Singleton.getInstance();  
  
        System.out.println(s1 == s2);  
    }  
}
```



```
Singleton Object Created  
true  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```

Exercise 2: Implementing the Factory Method Pattern

// java code

```
interface Shape {  
    void draw();  
}
```

```
class Circle implements Shape {  
    public void draw() {  
        System.out.println("Circle");  
    }  
}
```

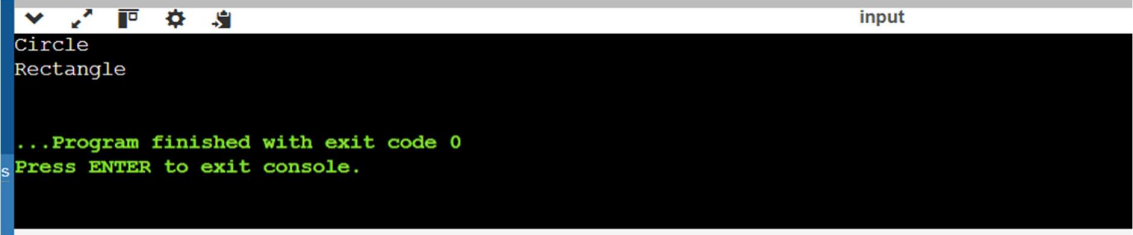
```
class Rectangle implements Shape {  
    public void draw() {  
        System.out.println("Rectangle");  
    }  
}
```

```
class ShapeFactory {  
    public Shape getShape(String type) {  
        if (type.equalsIgnoreCase("circle"))  
            return new Circle();  
        else  
            return new Rectangle();  
    }  
}
```

```
public class Main {  
    public static void main(String[] args) {  
        ShapeFactory factory = new ShapeFactory();  
    }  
}
```

```
Shape s1 = factory.getShape("circle");  
s1.draw();
```

```
Shape s2 = factory.getShape("rectangle");  
s2.draw();  
}  
}
```



A screenshot of a terminal window with a dark background. The title bar at the top is light gray and contains several icons on the left and the word "input" on the right. The terminal content shows the words "Circle" and "Rectangle" on separate lines. Below these, a green message states "...Program finished with exit code 0". At the bottom, a green prompt "s" is followed by the text "Press ENTER to exit console.".

```
Circle  
Rectangle  
  
...Program finished with exit code 0  
s Press ENTER to exit console.
```